

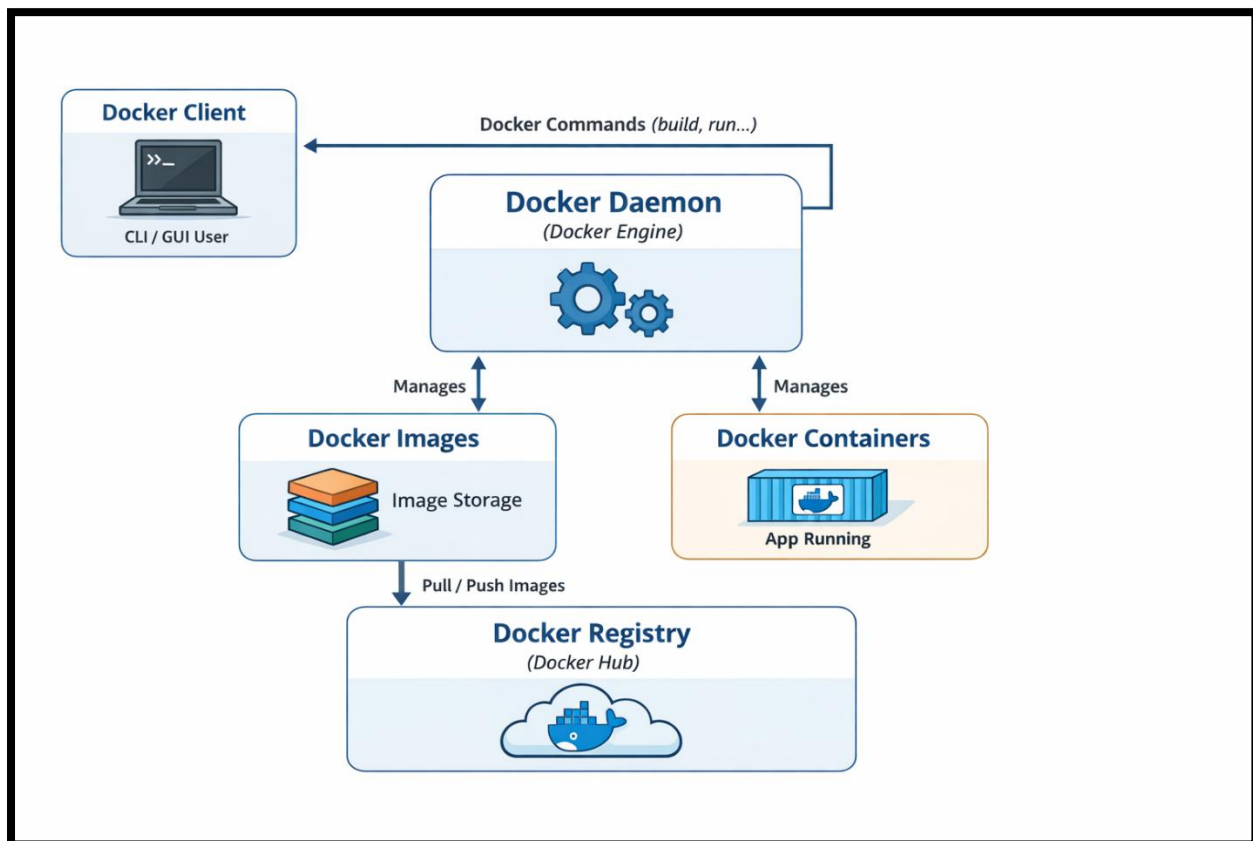
Experiment 9:

7. Laboratory Exercise

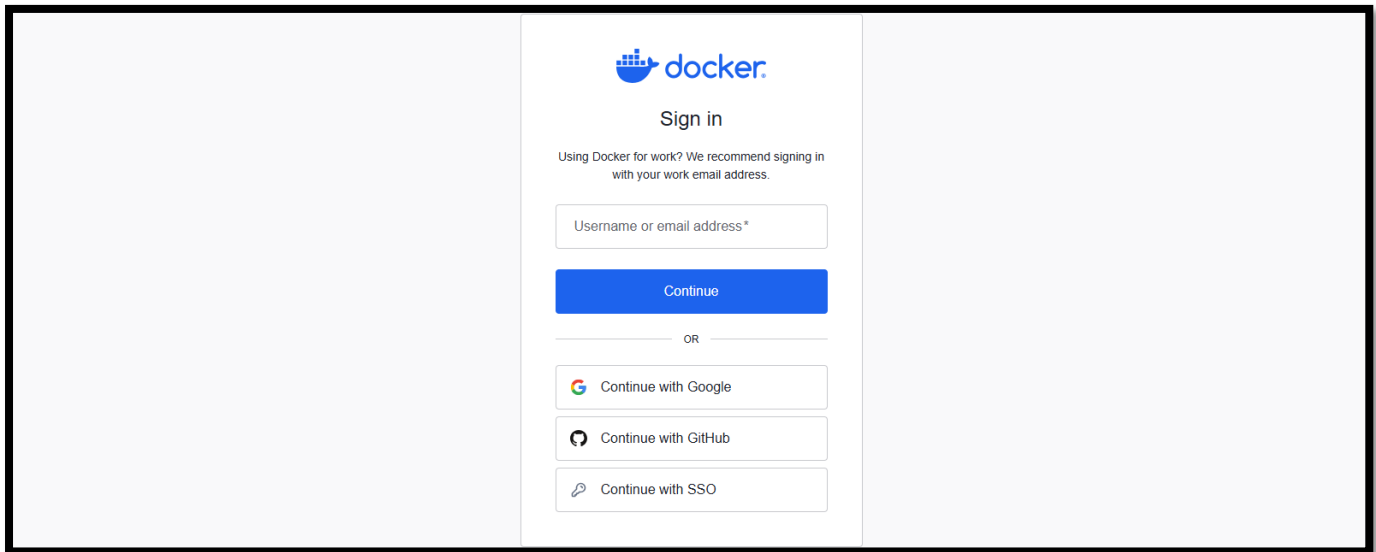
A. Procedure:

a. Answer the following:

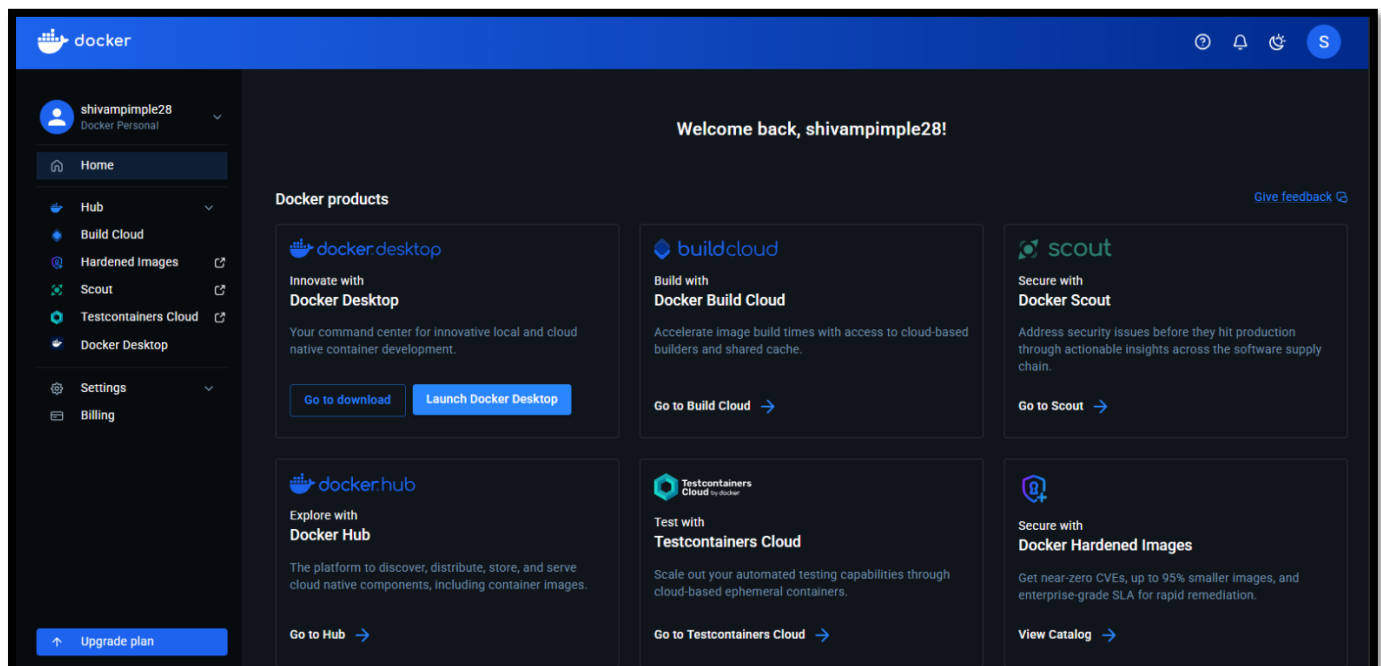
- Explain docker architecture with diagram.



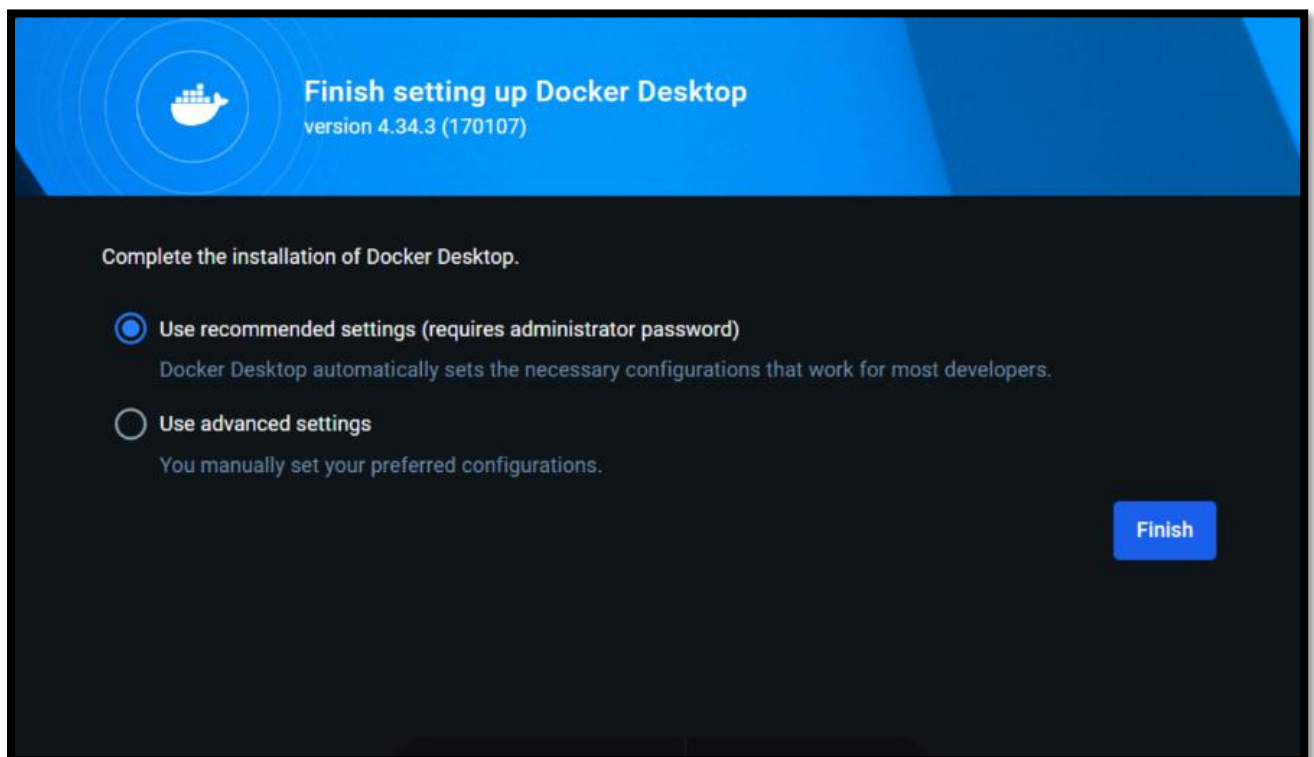
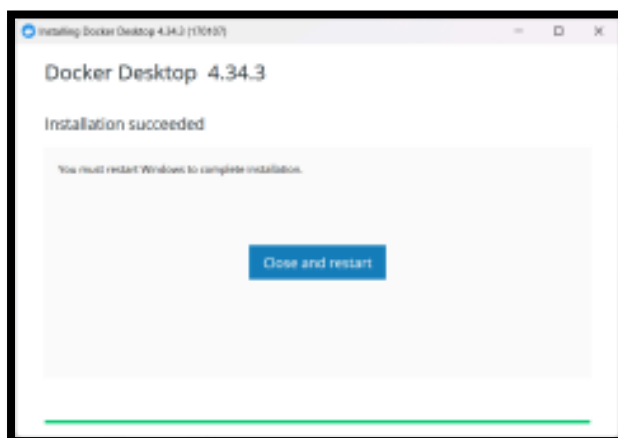
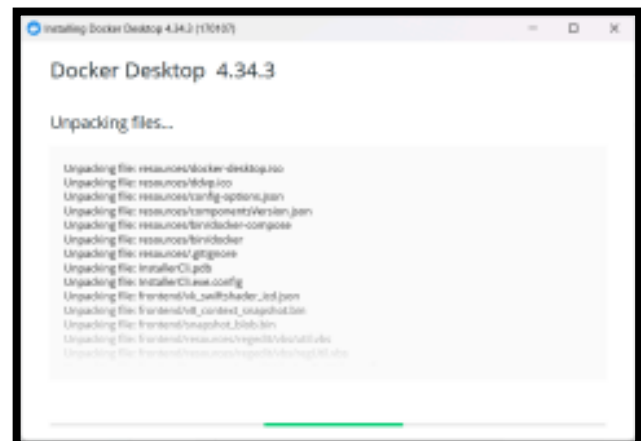
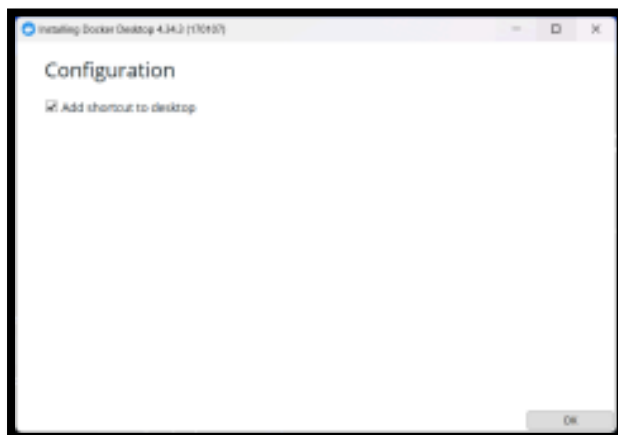
- **Create Docker Hub account – screenshot of steps related to account creation**



The screenshot shows the Docker 'Sign in' page. At the top is the Docker logo. Below it, the text 'Sign in' is centered. A message states: 'Using Docker for work? We recommend signing in with your work email address.' There is a text input field labeled 'Username or email address *'. Below the field is a blue 'Continue' button. Underneath is a horizontal line with 'OR' in the center. Below this are three more buttons: 'Continue with Google' (with the Google logo), 'Continue with GitHub' (with the GitHub logo), and 'Continue with SSO' (with a key icon).



- **Download and install Docker Desktop – screenshots of installation steps**



- Execute the following docker commands and take screenshots

1. Docker version

```
C:\Users\MOAD>Docker version
Client:
Version:      27.2.0
API version:  1.47
Go version:   go1.21.13
Git commit:   3ab4256
Built:        Tue Aug 27 14:17:17 2024
OS/Arch:      windows/amd64
Context:      desktop-linux

Server: Docker Desktop 4.34.3 (170107)
Engine:
Version:      27.2.0
API version:  1.47 (minimum version 1.24)
Go version:   go1.21.13
Git commit:   3ab5c7d
Built:        Tue Aug 27 14:15:15 2024
OS/Arch:      linux/amd64
Experimental: false
containerd:
Version:      1.7.20
GitCommit:    8fc6bcbff51318944179630522a095cc9dbf9f353
runc:
Version:      1.1.13
GitCommit:    v1.1.13-0-g58aa920
docker-init:
Version:      0.19.0
GitCommit:    de40ad0
```

```
C:\Users\MOAD>docker --version
Docker version 27.2.0, build 3ab4256

C:\Users\MOAD>
```

2. Docker login

```
C:\Users\MOAD>docker login
Authenticating with existing credentials...
Login Succeeded

C:\Users\MOAD>
```

3. Docker images

```
C:\Users\MOAD>docker images
REPOSITORY          TAG                 IMAGE ID            CREATED             SIZE
leron1/trialimage   new                 2025f9f8fb8b       3 days ago         117MB
alennnnnn/trialimage new                 48bc7abe89f1       4 days ago         117MB
ubuntu              devel              730382b4a53a       5 days ago         157MB
devanggg16/testimage 123                1a687896d732       6 days ago         117MB
<none>              <none>             91832dcd7bc5       3 weeks ago        157MB
ubuntu              latest             186072bba1b2       5 weeks ago        117MB
ubuntu              questing           4a9232cc47bf       3 months ago       135MB
docker/welcome-to-docker latest             c4d56c24da4f       8 months ago       22.2MB

C:\Users\MOAD>
```

4. Docker pull image

```
C:\Users\MOAD>docker pull ubuntu
Using default tag: latest
latest: Pulling from library/ubuntu
Digest: sha256:84e77dee7d1bc93fb029a45e3c6cb9d8aa4831ccfcc7103d36e876938d28895b
Status: Downloaded newer image for ubuntu:latest
docker.io/library/ubuntu:latest

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu

C:\Users\MOAD>|
```

```
C:\Users\MOAD>docker pull ubuntu:20.04
20.04: Pulling from library/ubuntu
13b7e930469f: Download complete
Digest: sha256:8feb4d8ca5354def3d8fce243717141ce31e2c428701f6682bd2fafa15388214
Status: Downloaded newer image for ubuntu:20.04
docker.io/library/ubuntu:20.04

What's next:
  View a summary of image vulnerabilities and recommendations → docker scout quickview ubuntu:20.04

C:\Users\MOAD>
```

5. Docker images help

```
C:\Users\MOAD>docker images --help

Usage:  docker images [OPTIONS] [REPOSITORY[:TAG]]

List images

Aliases:
  docker image ls, docker image list, docker images

Options:
  -a, --all                Show all images (default hides intermediate images)
  --digests               Show digests
  -f, --filter filter      Filter output based on conditions provided
  --format string          Format output using a custom template:
                           'table':          Print output in table format
                           with column headers (default)
                           'table TEMPLATE':  Print output in table format
                           using the given Go template
                           'json':           Print in JSON format
                           'TEMPLATE':       Print output using the given
                           Go template.
                           Refer to https://docs.docker.com/go/formatting/
                           for more information about formatting output with
                           templates
  --no-trunc              Don't truncate output
  -q, --quiet             Only show image IDs
  --tree                  List multi-platform images as a tree (EXPERIMENTAL)

C:\Users\MOAD>|
```

1. Docker images commands

```
C:\Users\MOAD>docker images -a
REPOSITORY          TAG             IMAGE ID        CREATED         SIZE
leroni/trialimage    new             2025f9f8fb8b   3 days ago     117MB
alennnnnn/trialimage new             48bc7abe89f1   4 days ago     117MB
ubuntu              devel          730382b4a53a   5 days ago     157MB
devanggg16/testimage 123            1a687896d732   6 days ago     117MB
<none>              <none>         91832dcd7bc5   3 weeks ago     157MB
ubuntu              latest         186072bba1b2   5 weeks ago     117MB
ubuntu              questing       4a9232cc47bf   3 months ago    135MB
docker/welcome-to-docker latest         c4d56c24da4f   8 months ago    22.2MB
ubuntu              20.04         8feb4d8ca535   12 months ago   109MB
```

```
C:\Users\MOAD>docker images -q
2025f9f8fb8b
48bc7abe89f1
730382b4a53a
1a687896d732
91832dcd7bc5
186072bba1b2
4a9232cc47bf
c4d56c24da4f
8feb4d8ca535
```

```
C:\Users\MOAD>docker images -f "dangling=true"
REPOSITORY TAG      IMAGE ID    CREATED     SIZE
<none>     <none>    91832dcd7bc5  3 weeks ago 157MB
```

```
C:\Users\MOAD>
```

```
C:\Users\MOAD>docker images -f "dangling=false" -q
2025f9f8fb8b
48bc7abe89f1
730382b4a53a
1a687896d732
186072bba1b2
4a9232cc47bf
c4d56c24da4f
8feb4d8ca535
```

```
C:\Users\MOAD>
```

2. Docker run commands

```
C:\Users\MOAD>docker run ubuntu
```

```
C:\Users\MOAD>docker ps -a
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS                               NAMES
ec7c1273c9f5   ubuntu        "/bin/bash"             29 seconds ago Exited (0) 29 seconds ago          quizzical_go
ldstine        39ccc3174029   paarthnginx            "nginx -g 'daemon of..." About an hour ago Exited (255) 7 minutes ago        0.0.0.0:8888->80/tcp charming_cha
tterjee        16f320921550   paarthnginx            "nginx -g 'daemon of..." About an hour ago Exited (0) About an hour ago      elegant_boyd
527783efee6b   kalilinux/kali-last-release "bash"                 25 hours ago   Exited (1) 25 hours ago            determined_h
amilton        eb0109fda54f   kalilinux/kali-last-release "bash"                 25 hours ago   Exited (130) 25 hours ago          adoring_shaw
4c257bd92d64   kalilinux/kali-last-release "bash"                 25 hours ago   Exited (0) 25 hours ago            laughing_pto
lemy           394f2bbe7b40   crishanginx            "nginx -g 'daemon of..." 28 hours ago   Exited (255) 25 hours ago          0.0.0.0:8888->80/tcp cool_matsumo
to             eba10fb82e25   65185a480078           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            pedantic_her
schel          e77ff8a5bb7a   ada7a41f9f94           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            tender_volha
rd             8b94a253ea2b   ada7a41f9f94           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            wonderful_ro
entgen         1900d731f95f   ab83045c1fba           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            modest_germa
in             d411e614ff13   ab83045c1fba           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            tender_shami
r              502f0fd17f81   ab83045c1fba           "nginx -g 'daemon of..." 28 hours ago   Exited (0) 28 hours ago            sharp_lampor
t              e5dd71aa6124   186072bba1b2           "/bin/bash"             2 days ago     Exited (0) 2 days ago              clever_brahm
agupta         0bd52abc82b3   186072bba1b2           "bash"                  7 days ago     Exited (255) 6 days ago            Paarth
b9173483316e   66460d557b25   "/bin/bash"             5 months ago   Exited (0) 5 months ago            charming_nas
h              dec01bc61541   66460d557b25           "/bin/bash"             5 months ago   Exited (0) 5 months ago            pensive_wesc
off
```

```
C:\Users\MOAD>
```

```
C:\Users\MOAD>docker run -it ubuntu
root@c8b6e2b84733:/# exit
exit
```

```
C:\Users\MOAD>docker run -it ubuntu
root@c8b6e2b84733:/# exit
exit

C:\Users\MOAD>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
1c2437477f09   ubuntu   "/bin/bash"   4 minutes ago   Up 4 minutes           intelligent_moser

C:\Users\MOAD>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
c8b6e2b84733   ubuntu   "/bin/bash"   About a minute ago   Exited (0) 30 seconds ago   adoring_hofstadter
1c2437477f09   ubuntu   "/bin/bash"   5 minutes ago   Up 5 minutes               intelligent_moser
336f5631abdc   ubuntu   "bash"       3 days ago   Exited (255) 47 minutes     go
c9ad063ec206   ubuntu   "bash"       4 days ago   Exited (255) 4 days ago     Alen
534d451ccb65   ubuntu   "bash"       6 days ago   Exited (255) 4 days ago     Devang
55e845277509   docker/welcome-to-docker:latest   "/docker-entrypoint..." 8 weeks ago   Exited (255) 6 days ago     0.0.0.0:8088->80/tcp welcome-to-docker

C:\Users\MOAD>
```

8. Docker ps

```
C:\Users\MOAD>docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
1c2437477f09   ubuntu   "/bin/bash"   4 minutes ago   Up 4 minutes           intelligent_moser

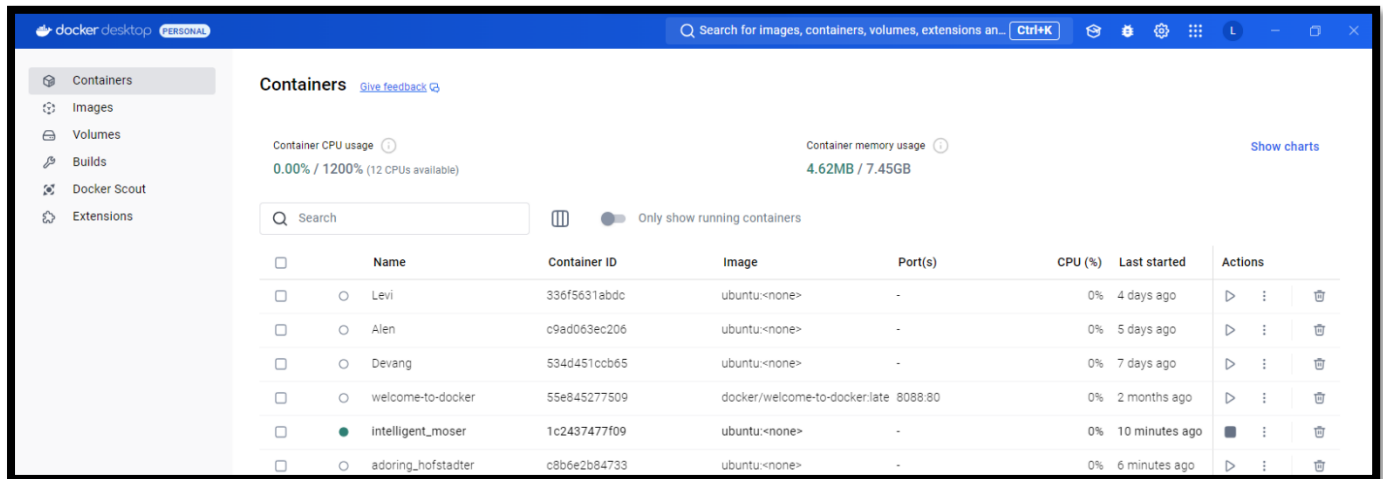
C:\Users\MOAD>docker ps -a
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS   NAMES
c8b6e2b84733   ubuntu   "/bin/bash"   About a minute ago   Exited (0) 30 seconds ago   adoring_hofstadter
1c2437477f09   ubuntu   "/bin/bash"   5 minutes ago   Up 5 minutes               intelligent_moser
336f5631abdc   ubuntu   "bash"       3 days ago   Exited (255) 47 minutes     go
c9ad063ec206   ubuntu   "bash"       4 days ago   Exited (255) 4 days ago     Alen
534d451ccb65   ubuntu   "bash"       6 days ago   Exited (255) 4 days ago     Devang
55e845277509   docker/welcome-to-docker:latest   "/docker-entrypoint..." 8 weeks ago   Exited (255) 6 days ago     0.0.0.0:8088->80/tcp welcome-to-docker

C:\Users\MOAD>
```

9. Docker start container

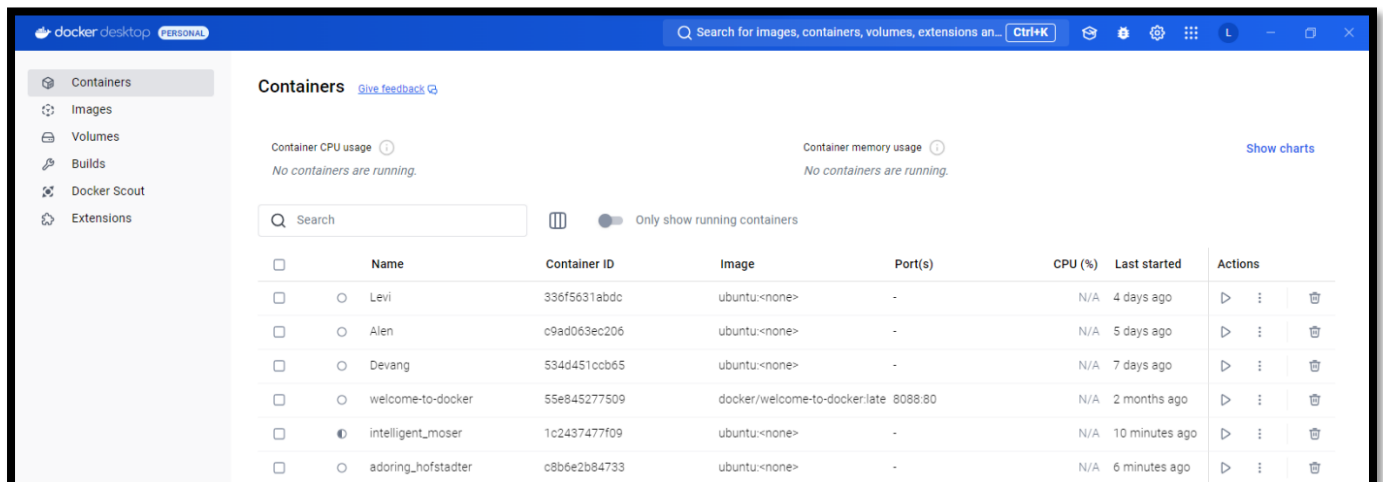
```
C:\Users\MOAD>docker start 1c2437477f09
1c2437477f09

C:\Users\MOAD>
```



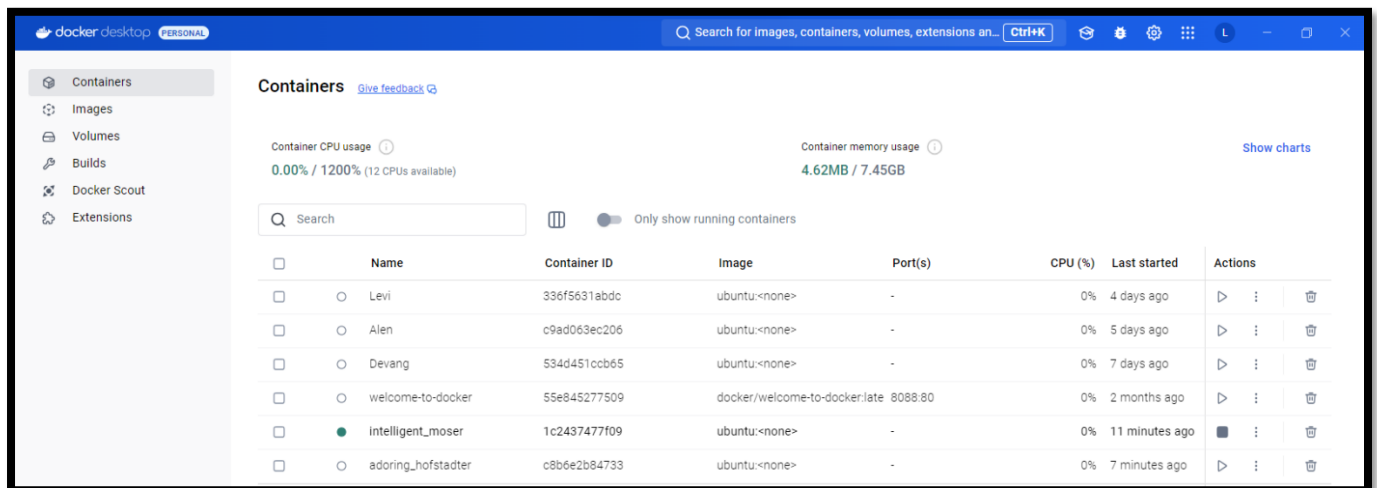
10. Docker pause container

```
C:\Users\MOAD>docker pause 1c2437477f09
1c2437477f09
C:\Users\MOAD>
```



11. Docker unpause container

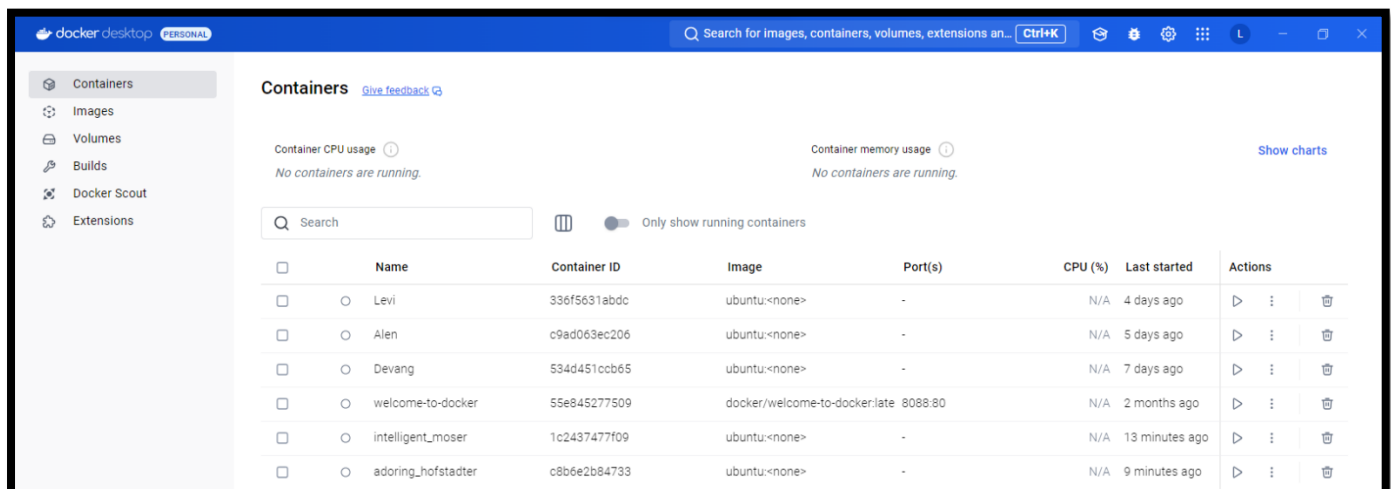
```
C:\Users\MOAD>docker unpause 1c2437477f09
1c2437477f09
C:\Users\MOAD>
```

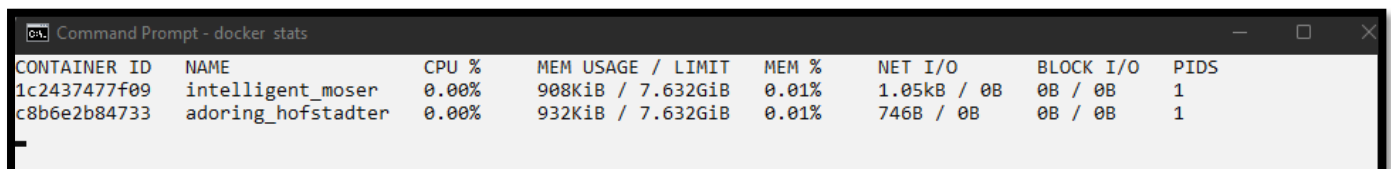
12. Docker stop container

```
C:\Users\MOAD>docker stop 1c2437477f09
1c2437477f09

C:\Users\MOAD>
```



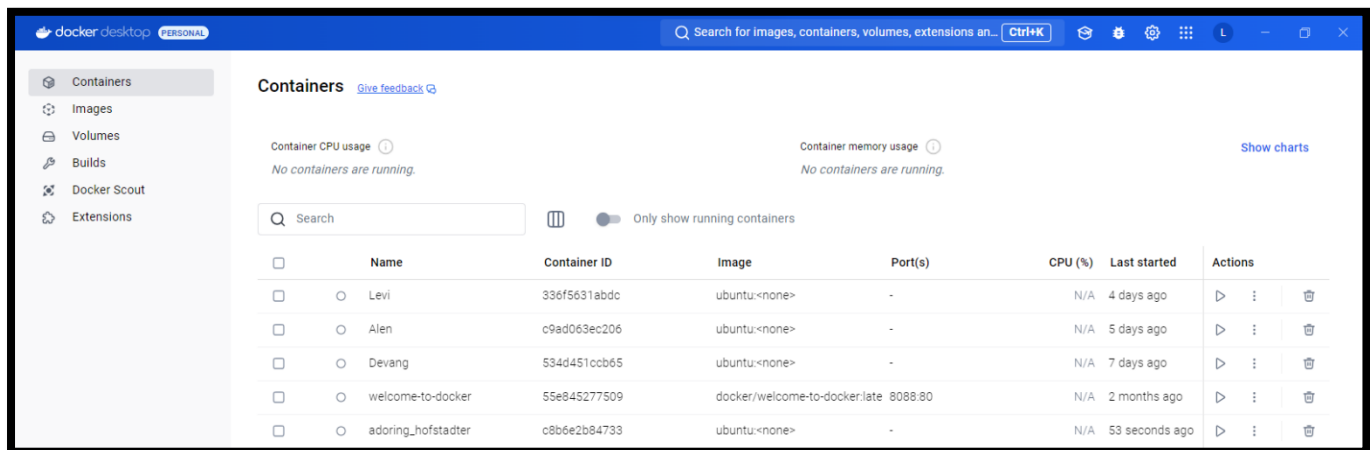
13. Docker stats container



14. Docker rm container

```
C:\Users\MOAD>docker rm 1c2437477f09
1c2437477f09

C:\Users\MOAD>
```



15. Docker inspect

```

C:\Users\MOAD>docker inspect ubuntu
[ ]
Error: No such object: ubuntu

C:\Users\MOAD>docker inspect ubuntu
[
  {
    "Id": "sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c",
    "RepoTags": [
      "ubuntu:latest"
    ],
    "RepoDigests": [
      "ubuntu@sha256:186072bba1b2f436cbb91ef2567abca677337cfc786c86e107d25b7072feef0c"
    ],
    "Parent": "",
    "Comment": "",
    "Created": "2026-02-23T17:17:56.132176015Z",
    "DockerVersion": "27.2.0",
    "Author": "",
    "Config": {
      "Hostname": "",
      "Domainname": "",
      "User": "",
      "AttachStdin": false,
      "AttachStdout": false,
      "AttachStderr": false,
      "Tty": false,
      "OpenStdin": false,
      "StdinOnce": false
    }
  }
]

```

16. Docker rmi

```

C:\Users\MOAD>docker images
REPOSITORY          TAG         IMAGE ID      CREATED      SIZE
leron1/trialimage   new        2025f9f8fb8b  3 days ago  117MB
alennnnnn/trialimage new        48bc7abe89f1  4 days ago  117MB
ubuntu              devel      730382b4a53a  5 days ago  157MB
devanggg16/testimage 123       1a687896d732  6 days ago  117MB
<none>              <none>    91832dcd7bc5  3 weeks ago 157MB
ubuntu              latest    186072bba1b2  5 weeks ago 117MB
ubuntu              questing  4a9232cc47bf  3 months ago 135MB
docker/welcome-to-docker latest     c4d56c24da4f  8 months ago 22.2MB
ubuntu              20.04    8feb4d8ca535  12 months ago 109MB

C:\Users\MOAD>docker rmi vdevanggg16/testimage:123
Error response from daemon: No such image: vdevanggg16/testimage:123

C:\Users\MOAD>docker rmi devanggg16/testimage:123
Untagged: devanggg16/testimage:123
Deleted: sha256:1a687896d7328e04ea9839bbaa72e26e1ebd25c23f6b9267acd9bbb0fb700d8c

```

```
C:\Users\MOAD>docker rmi devanggg16/testimage:123
Untagged: devanggg16/testimage:123
Deleted: sha256:1a687896d7328e04ea9839bbaa72e26e1ebd25c23f6b9267acd9bbb0fb700d8c

C:\Users\MOAD>docker images
```

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
leron1/trialimage	new	2025f9f8fb8b	3 days ago	117MB
alennnnnn/trialimage	new	48bc7abe89f1	4 days ago	117MB
ubuntu	devel	730382b4a53a	5 days ago	157MB
<none>	<none>	91832dcd7bc5	3 weeks ago	157MB
ubuntu	latest	186072bba1b2	5 weeks ago	117MB
ubuntu	questing	4a9232cc47bf	3 months ago	135MB
docker/welcome-to-docker	latest	c4d56c24da4f	8 months ago	22.2MB
ubuntu	20.04	8feb4d8ca535	12 months ago	109MB

17. Docker commit

```
C:\Users\MOAD>docker ps -a
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	POR
TS	NAMES				
ec7c1273c9f5	ubuntu	"/bin/bash"	19 minutes ago	Exited (0) 19 minutes ago	
39ccc3174029	paarthnginx	"nginx -g 'daemon of..."	2 hours ago	Exited (255) 26 minutes ago	0.0
.0.0:8888->80/tcp	charming_chatterjee				
16f320921550	paarthnginx	"nginx -g 'daemon of..."	2 hours ago	Exited (0) 2 hours ago	
	elegant_boyd				
527783efee6b	kalilinux/kali-last-release	"bash"	25 hours ago	Exited (1) 25 hours ago	
	determined_hamilton				
eb0109fda54f	kalilinux/kali-last-release	"bash"	25 hours ago	Exited (130) 25 hours ago	
	adoring_shaw				
4c257bd92d64	kalilinux/kali-last-release	"bash"	25 hours ago	Exited (0) 25 hours ago	
	laughing_ptolemy				
394f2bbe7b40	crishanginx	"nginx -g 'daemon of..."	28 hours ago	Exited (255) 25 hours ago	0.0
.0.0:8888->80/tcp	cool_matsumoto				
eba10fb82e25	65185a480078	"nginx -g 'daemon of..."	28 hours ago	Exited (0) 28 hours ago	
	pedantic_herschel				
e77ff8a5bb7a	ada7a41f9f94	"nginx -g 'daemon of..."	28 hours ago	Exited (0) 28 hours ago	
	tender_volhard				
8b94a253ea2b	ada7a41f9f94	"nginx -g 'daemon of..."	28 hours ago	Exited (0) 28 hours ago	
	wonderful_roentgen				
1900d731f95f	ab83045c1fba	"nginx -g 'daemon of..."	28 hours ago	Exited (0) 28 hours ago	

```
C:\Users\MOAD>docker commit 39ccc3174029 radhaursekar12/webmaster:1
sha256:45d41e26bbbed4bcd6b6de49f7ebf8ced3225da521067a5eb2ec0ce20ab77a3f5

C:\Users\MOAD>
```

18. Docker push

```
C:\Users\MOAD>docker push radhaursekar12/webmaster:1
The push refers to repository [docker.io/radhaursekar12/webmaster]
e09a1cadf2d6: Pushed
da5a52ccd7bc: Pushed
817807f3c64e: Mounted from paarth23/trailimage
4488e37f4b2d: Pushed
4577ba70524a: Pushed
088790e71b98: Pushed
1: digest: sha256:45d41e26bbbed4bcd6b6de49f7ebf8ced3225da521067a5eb2ec0ce20ab77a3f5 size: 1611
```

19. Docker history image

```
C:\Users\MOAD>docker images -q
29e960779256
2025f9f8fb8b
48bc7abe89f1
730382b4a53a
91832dcd7bc5
186072bba1b2
4a9232cc47bf
c4d56c24da4f
8feb4d8ca535

C:\Users\MOAD>docker history 29e960779256
IMAGE          CREATED        CREATED BY          SIZE      COMMENT
29e960779256   6 minutes ago /bin/bash          12.3kB
186072bba1b2   5 weeks ago   /bin/sh -c #(nop)  CMD ["/bin/bash"]   0B
<missing>       5 weeks ago   /bin/sh -c #(nop)  ADD file:3f78aa860931e0853... 87.6MB
<missing>       5 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers.... 0B
<missing>       5 weeks ago   /bin/sh -c #(nop)  LABEL org.opencontainers.... 0B
<missing>       5 weeks ago   /bin/sh -c #(nop)  ARG LAUNCHPAD_BUILD_ARCH    0B
<missing>       5 weeks ago   /bin/sh -c #(nop)  ARG RELEASE                 0B

C:\Users\MOAD>
```